

Project Title: Loan Origination & Credit Risk Management System

Business Domain: Banking & Financial Services

1. Overview

The Loan Origination & Credit Risk Management System automates the end-to-end lifecycle of retail and commercial lending operations, including customer onboarding, loan application processing, credit scoring, disbursement, repayment tracking, and collections. Built using an MVC architecture, it supports both Java (Spring MVC) and .NET (ASP.NET Core MVC) frameworks.

The system consists of five key modules:

1. Customer Onboarding & KYC Verification
2. Loan Application & Eligibility Assessment
3. Credit Scoring & Risk Evaluation
4. Loan Disbursement & Repayment Tracking
5. Collections & Delinquency Management

2. Assumptions

- **Database:** MySQL or SQL Server.
- **Roles:** Loan Officer, Credit Analyst, Underwriter, Collections Agent, Admin.
- **ORM:** Hibernate (Java), Entity Framework Core (.NET).
- KYC document verification uses mock OCR services.
- Credit bureau scores are fetched via mock APIs.

3. Module-Level Design

3.1 Customer Onboarding & KYC Verification Module

Purpose: Manages prospective borrower registration and identity verification.

Controller:

- CustomerController
 - registerCustomer()
 - uploadKycDocument()

- verifyKycStatus()
- getCustomerProfile()

Service:

- CustomerService
 - Handles customer data capture, KYC validation, and AML screening.

Model:

- Customer Entity
 - Attributes:
 - customerId (PK)
 - fullName
 - panNumber
 - aadhaarNumber
 - kycStatus (PENDING, VERIFIED, REJECTED)
 - registrationDate

3.2 Loan Application & Eligibility Assessment Module

Purpose: Captures loan requests and evaluates initial eligibility.

Controller:

- LoanApplicationController
 - submitLoanApplication()
 - checkEligibility()
 - getApplicationStatus()

Service:

- LoanApplicationService
 - Implements business rules for income, employment, and existing liability checks.

Model:

- LoanApplication Entity

- Attributes:
 - applicationId (PK)
 - customerId (FK)
 - loanType
 - requestedAmount
 - tenureMonths
 - applicationStatus (SUBMITTED, UNDER_REVIEW, APPROVED, REJECTED)

3.3 Credit Scoring & Risk Evaluation Module

Purpose: Calculates credit scores and assigns risk grades to applicants.

Controller:

- CreditScoreController
 - fetchBureauScore()
 - calculateInternalScore()
 - assignRiskGrade()

Service:

- CreditScoreService
 - Combines bureau data, repayment history, and internal scoring models.

Model:

- CreditScore Entity
 - Attributes:
 - scoreId (PK)
 - customerId (FK)
 - bureauScore
 - internalScore
 - riskGrade (LOW, MEDIUM, HIGH)
 - evaluationDate

3.4 Loan Disbursement & Repayment Tracking Module

Purpose: Processes approved loan disbursements and tracks EMI repayments.

Controller:

- DisbursementController
 - disburseLoan()
 - recordRepayment()
 - getRepaymentSchedule()

Service:

- DisbursementService
 - Generates amortization schedules and posts repayment transactions.

Model:

- Repayment Entity
 - Attributes:
 - repaymentId (PK)
 - applicationId (FK)
 - dueDate
 - emiAmount
 - paidAmount
 - repaymentStatus (PENDING, PAID, OVERDUE)

3.5 Collections & Delinquency Management Module

Purpose: Tracks overdue accounts and coordinates recovery actions.

Controller:

- CollectionsController
 - flagDelinquentAccount()
 - assignToAgent()
 - logRecoveryAction()

Service:

- CollectionsService
 - Manages bucket aging, agent assignment, and escalation workflows.

Model:

- CollectionsLog Entity
 - Attributes:
 - logId (PK)
 - applicationId (FK)
 - daysPastDue
 - actionTaken
 - actionDate
 - recoveryStatus (PENDING, RECOVERED, WRITTEN_OFF)

4. Database Schema

Customer Table

```
CREATE TABLE Customer (  
  customerId INT AUTO_INCREMENT PRIMARY KEY,  
  fullName VARCHAR(100),  
  panNumber VARCHAR(20),  
  aadhaarNumber VARCHAR(20),  
  kycStatus ENUM('PENDING', 'VERIFIED', 'REJECTED'),  
  registrationDate DATE  
);
```

LoanApplication Table

```
CREATE TABLE LoanApplication (  
  applicationId INT AUTO_INCREMENT PRIMARY KEY,  
  customerId INT,  
  loanType VARCHAR(50),  
  requestedAmount DECIMAL(12,2),  
  tenureMonths INT,  
  applicationStatus ENUM('SUBMITTED', 'UNDER_REVIEW', 'APPROVED', 'REJECTED'),  
  FOREIGN KEY (customerId) REFERENCES Customer(customerId)  
);
```

CreditScore Table

```
CREATE TABLE CreditScore (  
  scoreId INT AUTO_INCREMENT PRIMARY KEY,
```

```
customerId INT,  
bureauScore INT,  
internalScore INT,  
riskGrade ENUM('LOW', 'MEDIUM', 'HIGH'),  
evaluationDate DATE,  
FOREIGN KEY (customerId) REFERENCES Customer(customerId)  
);
```

Repayment Table

```
CREATE TABLE Repayment (  
repaymentId INT AUTO_INCREMENT PRIMARY KEY,  
applicationId INT,  
dueDate DATE,  
emiAmount DECIMAL(12,2),  
paidAmount DECIMAL(12,2),  
repaymentStatus ENUM('PENDING', 'PAID', 'OVERDUE'),  
FOREIGN KEY (applicationId) REFERENCES LoanApplication(applicationId)  
);
```

CollectionsLog Table

```
CREATE TABLE CollectionsLog (  
logId INT AUTO_INCREMENT PRIMARY KEY,  
applicationId INT,  
daysPastDue INT,  
actionTaken VARCHAR(255),  
actionDate DATE,  
recoveryStatus ENUM('PENDING', 'RECOVERED', 'WRITTEN_OFF'),  
FOREIGN KEY (applicationId) REFERENCES LoanApplication(applicationId)  
);
```

5. Local Deployment Details

- Install DB + JDK 17/21/24 or .NET SDK 8.0
- Deploy using Tomcat or Kestrel

Steps:

1. Clone repository.
2. Run SQL schema.
3. Configure app settings files.
4. Build & run the project.

6. Conclusion

The Loan Origination & Credit Risk Management System provides a robust MVC-based structure for lending operations, credit evaluation, and recovery management. It is ideal for training scenarios and local deployment.